SYS1000 System Development – Exam Project Spring of 2026.

Project Overview:
This semester, students will design and implement a system similar to Airbnb. The project involves creating a web-based platform that allows homeowners to list their properties for rent and enables clients to search, filter, and book these properties.

Company Background:
Our fictional company, **Home Rentals**, is a local startup aiming to revolutionize the rental market by providing a seamless platform for homeowners and clients. Homeowners can list properties of various types, sizes, and locations, while clients can browse and book these properties based on their specific needs.

Project Requirements:
As part of this project, students are expected to:

- Develop a comprehensive frontend solution.
- Design an appropriate database structure to support the system's operations.

Note that no interaction between the frontend and backend is required for this project, as backend development knowledge is not assumed.

Tasks for Students:

1. **Identify Functional Requirements:**
    - Determine what features the system needs to offer for both homeowners and clients.
    - Consider functionalities such as user registration, property listings, search and filtering, booking, and reviews.
2. **Identify Non-Functional Requirements:**
    - Consider aspects such as performance, security, scalability, and usability.
    - Think about how these requirements will impact the design of your system.
3. **Design the Database Structure:**
    - Create a schema that supports the functionality of your application.
    - Think about what data needs to be stored and how it will be organized.
4. **Develop the Frontend Interface:**
    - Design a user-friendly interface that allows users to interact with the system effectively.

Deliverables:

- A written report detailing the identified functional and non-functional requirements.
- A database schema design.
- A frontend prototype demonstrating the key functionalities of the system.

Conclusion:
This project aims to provide students with hands-on experience in building a complex web-based system that addresses real-world needs. By completing this project, students will gain valuable skills in system design, database management, and frontend development, preparing them for future challenges in the field of system development.

Formal requirements

The exam project is carried out in groups of 4, 5 or 6 members, according to group composition. Submission takes place in WISEflow, and the submission date will be published together with the ordinary exam plan for spring 2025.

The report must contain (minimum requirements):

- Project organization, overall project plan
- Overall Problem Description/Initial Problem Identification
- List of requirements
- Use case (diagram and descriptions) with DFD fragments
- Data Catalog
- Conceptual and logical data model
- Prototype and screenshot draft, menu system/navigation
- Overall test plan
- Overall adoption strategy
- The development process
    - Actual implementation assessed against the chosen development model/method and whether the chosen development model/method was appropriate for this project
    - Deviations from the project plan during project implementation
    - History of project meetings (date, class, attendance, purpose of the meeting)
    - Any tools we have used in the project
- Conclusion
    - "What have we done?", "What have we learned?", "What could we have done differently?"

The report must have:

- Front Page
    - With candidate number, and place/time and part of the study programme
- Table of contents
    - All main and sub-chapters with page reference
- Summary or Abstract
    - A brief description of the structure of the report, i.e. what the chapters are about and how they are connected
- Introduction
    - What is this project about?
- Main chapters
    - Cf. minimum content requirements
    - Give the chapters good names
- Bibliography
    - Reference requirements: APA 7 (both for bibliography and source referencing in running text)
- Times New Roman 12 and line spacing 1.5
- The report is delivered as a Word document